MTH302

LESSON (41-45)

50 MCQS

20 Shorts Questions

10 Long Questions

Vu Nexus

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MCQs on Normal Distribution

1. What does the standard normal distribution represent?

A) A normal distribution with mean = 0 and variance = 1

B) A normal distribution with mean = 1 and variance = 0

C) A normal distribution with mean = 1 and variance = 1

D) A distribution with no variance

Answer: A

2. What shape does the curve of a normal distribution take?

A) Uniform

B) Bell-shaped

C) Skewed

D) Linear

Answer: B

3. In a normal distribution, what is the total area under the curve?

A) 0.5

B) 1

C) Infinity

D) 100

Answer: B

4. If $X \sim N(45, 0.000625)$, what is the standard deviation (σ)?

A) 45

B) 0.025

C) 0.000625

D) D) 0.1

Answer: B

5. What transformation converts a normal random variable $X \sim N(\mu, \sigma^2)$ to the standard normal distribution?

A) $Z = \frac{X\sigma}{\mu}$

B) $Z = X + \mu$

C) $Z = \mu \times X$

D) D) $Z = X^2$

Answer: A

6. What is the probability of Z lying between 0 and 1.9 in the standard normal distribution?

A) 0.4713

B) 0.5

C) 0.9772

D) 0.3413

Answer: A

7. Which of the following is a property of the standard normal distribution?

A) Skewed to the left

B) Symmetrical about the mean

C) Area greater than 1

D) Discrete values

Answer: B

8. For $Z > 2$, what is the probability?

A) 0.0228

B) 0.4772

C) 0.9772

D) 0.5

Answer: A

9. What is the probability that $Z < 1.96$ in the standard normal distribution?

A) 0.5

B) 0.975

C) 0.025

D) 0.4772

Answer: B

10. In a standard normal table, the value 1.96 corresponds to which confidence interval?

A) 90%

B) 95%

C) 99%

D) 50%

Answer: B

11. The Z-score is used to...

A) Convert any normal distribution to a standard normal distribution

B) Determine the mode of the distribution

C) Calculate variance directly

D) Find discrete probabilities

Answer: A

12.What is the formula for the probability density function of a normal distribution?

A) $y = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x2-\sigma\mu2)2}$

B) $y = e^{x-\mu}$

C) $y = x^2 + \mu$

D) $y = x1$

Answer: A

13. What is the area under the curve for $Z > 3$?

A) Approximately 0.0013

B) 0.5

C) 0.02
D) 0.9772
Answer: A

14. What is the significance of symmetry in the standard normal curve?
A) All probabilities are equal
B) Areas on both sides of the mean are identical
C) Only positive values are allowed
D) Negative values are ignored

Answer: B

15. If $X \sim N(50,16)$, the standard deviation (σ) is...
A) 16
B) 50
C) 4
D) 2
Answer: C

16. The 99% confidence interval for Z is approximately...
A) ± 1.96
B) ± 2.58
C) ± 3.00
D) ± 1.64
Answer: B

17. If $P(Z > 1.5) = 0.0668$, what is $P(Z < -1.5)$?
A) 0.0668
B) 0.5
C) 0.4332
D) 0.9332
Answer: A

18. What is the cumulative probability for $Z = 0$?
A) 0
B) 0.5
C) 1
D) Undefined

Answer: B

19. What is the probability $P(-2 < Z < 2)$?
A) 0.4772
B) 0.8185
C) 0.95
D) 0.3413
Answer: C

20. What does a Z-score of 2 signify?
A) The value is 2 standard deviations above the mean
B) The value is 2 units above the mean

C) The value is twice the variance
D) The value is equal to the mean

Answer: A

21. What is the probability that $Z < -2$?
A) 0.0228
B) 0.9772
C) 0.5
D) 0.3413
Answer: A

22. For $P(Z > 1.8)$, what is the probability?
A) 0.0359
B) 0.5
C) 0.4641
D) 0.9641
Answer: A

23. In a normal distribution, the probability of $Z > 0$ is...
A) 0.5
B) 1
C) 0
D) 0.25
Answer: A

24. If $P(Z > 3) \approx 0.0013$, then $P(Z < -3)$ is...
A) 0.0013
B) 0.5
C) 0.9987
D) 0.25
Answer: A

25. What is $P(1 < Z < 2)$?
A) 0.1359
B) 0.4772
C) 0.3413
D) 0.5
Answer: A

26. For $X \sim N(100,25)$, what is the Z-score for $X = 105$?
A) 1
B) 2
C) -1
D) 0.5
Answer: A

27. If $P(Z < -1)$ is 0.1587, what is $P(Z > 1)$?
A) 0.1587
B) 0.8413
C) 0.5

D) 0.025

Answer: A

28. What is the significance of the Z-score?

A) Measures the distance from the mean in standard deviations

B) Calculates mean

C) Identifies skewness

D) Determines mode

Answer: A

29. What percentage of values in a normal distribution lie within 1 standard deviation of the mean?

A) 68%

B) 95%

C) 99%

D) 50%

Answer: A

30. What is the area under the curve for $Z = 2.33$?

A) 0.4901

B) 0.9772

C) 0.4332

D) 0.5

Answer: B

31. What is the probability $P(-1 < Z < 1)$?

A) 0.6826

B) 0.4772

C) 0.3413

D) 0.8185

Answer: A

32. The Z-score for $X = 50$, given $X \sim N(40,100)$, is...

A) 1.0

B) 0.5

C) -1.0

D) 2.0

Answer: A

33. What is the cumulative probability for $Z = -2.33$?

A) 0.0099

B) 0.01

C) 0.0001

D) 0.9901

Answer: A

34. If $Z = -1.645$ corresponds to a one-tailed confidence level, what is the level?

A) 95%

B) 90%

C) 99%

D) 80%

Answer: B

35. How do you find the probability $P(-Z_1 < Z < Z_1)$ for a confidence interval?

A) Subtract the area outside the interval from 1

B) Add the areas outside the interval

C) Use only the mean

D) Double the Z-score

Answer: A

36. Which of the following describes a 99% confidence interval?

A) $Z = 1.96$

B) $Z = 2.33$

C) $Z = 2.58$

D) $Z = 1.64$

Answer: C

37. What is $P(Z > 1.96)$ in the standard normal distribution?

A) 0.025

B) 0.05

C) 0.975

D) 0.005

Answer: A

38. If $Z = 2.5$, what is the cumulative area under the curve?

A) 0.9938

B) 0.5

C) 0.0062

D) 1.0000

Answer: A

39. The cumulative area at $Z = -1.0$ is...

A) 0.1587

B) 0.8413

C) 0.5

D) 0.025

Answer: A

40. What is the probability of $P(Z < 3.0)$?

A) 0.9987

B) 0.0013

C) 0.5000

D) 0.9772

Answer: A

41. Linear interpolation is used in Z-tables to...

A) Estimate probabilities between tabulated Z-values

B) Calculate mean values

C) Find skewness
D) Define standard deviation

Answer: A

42. For $P(1.5 < Z < 1.8)$, what is the area?
A) 0.0309
B) 0.4332
C) 0.4641
D) D) 0.5
Answer: A

43. What does $P(Z > -1.5)$ equal?
A) 0.9332
B) 0.0668
C) 0.5
D) 0.3413
Answer: A

44. What is the value of Z for a cumulative probability of 0.975?
A) 1.96
B) 2.0
C) 1.64
D) 2.33
Answer: A

45. The Z-score for a value 3 standard deviations above the mean is...
A) 3
B) 1
C) 0
D) -3
Answer: A

46. What is the complement of $P(Z > 2.5)$?
A) $P(Z \leq 2.5)$
B) $P(Z < 2.5)$
C) 1 - $P(Z > 2.5)$
D) All of the above

Answer: D

47. In a two-tailed test with $\alpha = 0.05$, what are the critical Z-values?
A) ± 1.96
B) ± 2.33
C) ± 1.64
D) ± 2.58
Answer: A

48. For $X \sim N(60,36)$, the probability of $X < 54$ is calculated by finding...
A) Z and the cumulative probability
B) The mean

C) The mode
D) The standard deviation

Answer: A

49. If $P(Z < -1.28) = 0.1$, what is $P(Z > 1.28)$?
A) 0.1
B) 0.2
C) 0.5
D) 0.8
Answer: A

50. For a confidence level of 95%, the interval about the mean corresponds to which Z-scores?
A) ±1.96
B) ±1.64
C) ±2.33
D) ±2.58
Answer: A

Short Questions

1. **What is the mean and variance of the standard normal distribution?**

   Answer: Mean = 0, Variance = 1

2. **Write the formula for converting a random variable $X \sim N(\mu, \sigma^2)$ to a standard normal variable Z.**

Answer: $Z = \frac{X - \mu}{\sigma}$

3. **What is the shape of a normal distribution curve?**

   Answer: Bell-shaped

4. **What does the total area under the normal distribution curve represent?**

   Answer: The total probability, which equals 1

5. **Define the probability density function of a normal distribution.**

Answer: $f(x) = \sigma 12\pi e^{-\sqrt{(x2-\sigma\mu2)2}}$

6. **Explain the significance of $Z = 0$ in the standard normal distribution.**

   Answer: $Z = 0$ represents the mean of the standard normal distribution.

7. **How do you find the probability that $Z > z_1$ using a standard normal table?**

   Answer: Subtract the cumulative probability of $Z \leq z_1$ from 1.

8. **What is the probability of Z lying between -1.96 and 1.96?**

   Answer: Approximately 95%

9. **What does the cumulative probability for Z represent?**

Answer: It represents the probability that Z is less than or equal to a specific value.

10. **How do we use symmetry in the standard normal curve to simplify probability calculations?**

Answer: Probabilities for $Z > z_1$ and $Z < -z_1$ are equal due to symmetry.

11. **Write the formula for the standard normal distribution's probability density function.**

Answer: $f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$

12. **How is a 95% confidence interval for the standard normal distribution determined?**

Answer: It is the interval $-1.96 < Z < 1.96$, capturing 95% of the area under the curve.

13. **What is the Z-score if a value is 2 standard deviations above the mean?**

Answer: $Z = 2$

14. **Why is interpolation used in standard normal tables?**

Answer: To estimate probabilities for Z-values not directly listed in the table.

15. **What is the cumulative probability of $Z = 3$?**

Answer: Approximately 0.9987

16. **Explain the relationship between mean, variance, and standard deviation in a normaldistribution.**

Answer: Variance is the square of the standard deviation, and the mean is the central point of the distribution.

17. **If $X \sim N(100,25)$, what is the Z-score for $X = 105$?**

Answer: $Z = \frac{105 - 100}{5} = 1$

18. **What does it mean if $P(Z > 1.8) = 0.0359$?**

Answer: There is a 3.59% chance that Z will be greater than 1.8.

19. **Why are confidence intervals wider for higher confidence levels?**

Answer: Higher confidence levels require capturing more of the distribution, thus increasing the range.

20. **What does $P(-Z < Z < Z)$ represent in a normal distribution?**

Answer: It represents the probability of Z lying symmetrically within a specified range around the mean.

Long Questions

1. Explain the standard normal distribution and its key properties.

Answer:

The standard normal distribution is a specific type of normal distribution where the mean is 0 and the variance is 1 ($N(0,1)$). Its probability density function (PDF) is given by:

$$f(z) = e^{-\frac{z^2}{2}} \qquad \frac{1}{\sqrt{2\pi}}$$

Key properties:

- • The curve is bell-shaped and symmetrical about the mean ($z = 0$).
- • The total area under the curve is 1.
- Approximately 68% of the data lies within 1 standard deviation, 95% within 2, and 99.7% within 3 standard deviations from the mean.

- The distribution is useful for standardizing any normal distribution using the Z-score formula.

2. Derive the transformation formula to convert $X \sim N(\mu,\sigma^2)$ to the standard normal variable Z

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Answer:

Let $X \sim N(\mu,\sigma^2)$. The Z-score is a measure of how many standard deviations a particular value x is from the mean μ. The formula for transformation is:

$$Z = \frac{X - \mu}{\sigma}$$

Steps:

1. Subtract the mean (μ) from X to center the distribution at 0.
2. Divide by the standard deviation (σ) to normalize the variance to 1. This transformation ensures that $Z \sim N(0,1)$.

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3. How do you calculate probabilities using the standard normal distribution table? Give anexample.

Answer:

The standard normal table provides cumulative probabilities for Z values (i.e., $P(Z \leq z)$). Steps:

1. Locate the row corresponding to the first two digits of z (e.g., for $z = 1.23$, use "1.2").
2. Find the column matching the second decimal place (e.g., for $z = 1.23$, use "0.03").
3. The intersection gives the cumulative probability. Example: Find $P(Z \leq 1.23)$.

- From the table, $P(Z \leq 1.23) = 0.8907$.
- Interpretation: 89.07% of values are below $Z = 1.23$.

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4. What is the significance of confidence intervals in the context of normal distributions?Answer:

Confidence intervals (CIs) estimate the range within which a population parameter (e.g., mean) is likely to lie with a specified probability. For a normal distribution:

- A 95% CI corresponds to Z-values between -1.96 and 1.96.
- A 99% CI corresponds to Z-values between -2.58 and 2.58. The

formula for CI is:

$$CI = \mu \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where $Z_{\alpha/2}$ is the critical Z-value, σ is the standard deviation, and n is the sample size.

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5. Illustrate the use of symmetry in the standard normal distribution with an example.

Answer:

The standard normal distribution is symmetric about the mean ($Z = 0$). Example: Find $P(Z > 1.5)$.

- By symmetry, $P(Z > 1.5) = P(Z < -1.5)$.
- From the table, $P(Z < -1.5) = 1 - P(Z \leq 1.5) = 1 - 0.9332 = 0.0668$.
- Thus, $P(Z > 1.5) = 0.0668$.
  This property simplifies calculations by avoiding redundant table lookups.

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6. How is interpolation used in standard normal tables? Provide an example.Answer:

Interpolation is used to estimate probabilities for Z-values not directly listed in the table. Example: Find $P(Z = 1.965)$.

- From the table, $P(Z = 1.96) = 0.4750$ and $P(Z = 1.97) = 0.4756$.
- Using linear interpolation:

$P(Z = 1.965) = P(Z = 1.96) + 0.5 \cdot (P(Z = 1.97) - P(Z = 1.96)) = 0.4750 + 0.5 \cdot (0.4756 - 0$

- Estimated probability is 0.4753.

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7. What are the steps to calculate $P(-2 < Z < 1.5)$ using the standard normal distribution? Answer:

1. Find $P(Z < 1.5)$: From the table, $P(Z < 1.5) = 0.9332$.
2. Find $P(Z < -2)$: By symmetry, $P(Z < -2) = 1 - P(Z \leq 2) = 1 - 0.9772 = 0.0228$.
3. Subtract $P(Z < -2)$ from $P(Z < 1.5)$:

$$P(-2 < Z < 1.5) = P(Z < 1.5) - P(Z < -2) = 0.9332 - 0.0228 = 0.9104$$

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8. Explain the concept of the Z-score and its importance in statistics.

Answer:

The Z-score measures how many standard deviations a value is from the mean. It is calculated as:

$$Z = \frac{X - \mu}{\sigma}$$

Importance:

- Standardizes data for comparison across different distributions.
- Identifies outliers (values far from the mean).
- Facilitates probability calculations using the standard normal distribution.

  Example: For $X = 105$, $\mu = 100$, and $\sigma = 5$, $Z = \frac{105-100}{5} = 1$.

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9. Describe the relationship between the area under the standard normal curve and probabilities.

Answer:

The area under the standard normal curve represents cumulative probabilities:

- Total area = 1 (entire probability space).

- $P(Z < z)$: Area to the left of z.
- $P(Z > z)$: Area to the right of z, equal to $1 - P(Z \leq z)$.
- Symmetry ensures $P(Z < 0) = P(Z > 0) = 0.5$.

  Example: $P(Z > 2) = 1 - P(Z \leq 2) = 1 - 0.9772 = 0.0228$.

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10. How do you interpret probabilities in the context of normal distributions? Provide an example.

Answer:

Probabilities indicate the likelihood of a random variable falling within a certain range. Example: In $N(50,25)$, find $P(45 < X < 55)$:

1. Transform to Z:
   - $Z = \frac{45-50}{5} = -1, Z = \frac{55-50}{5} = 1$.
2. Use the standard normal table:
   - $P(-1 < Z < 1) = P(Z < 1) - P(Z < -1) = 0.8413 - 0.1587 = 0.6826$.
   - Probability is 68.26%.
